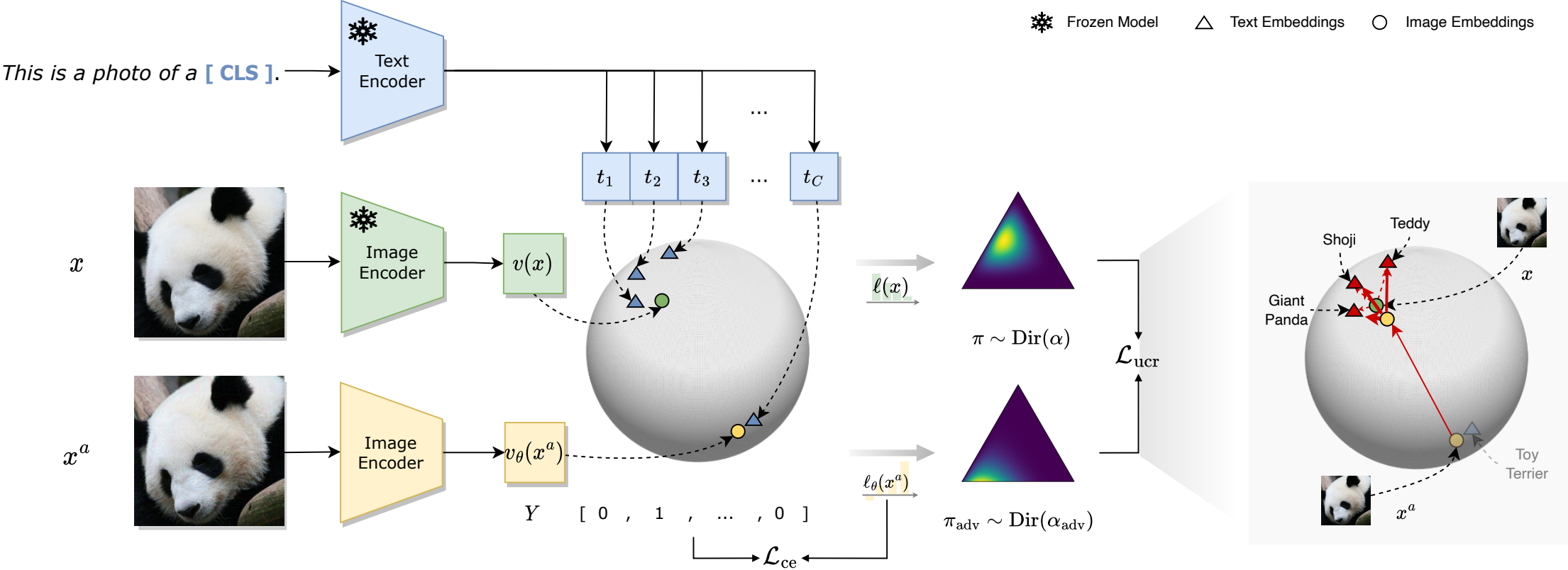
Frozen Model
Text Embeddings
Image Embeddings
This is a photo of a [CLS].
Text Encoder
t_1 t_2 t_3 ... t_C
x
Image Encoder
$v(x)$
x^a
Image Encoder
$v_\theta(x^a)$
Y [0 , 1 , ... , 0]
$\ell(x)$
$\ell_\theta(x^a)$
$\pi \sim \mathrm{Dir}(\alpha)$
$\pi_{\mathrm{adv}} \sim \mathrm{Dir}(\alpha_{\mathrm{adv}})$
$\mathcal{L}_{\mathrm{ucr}}$
$\mathcal{L}_{\mathrm{ce}}$
Teddy
Shoji
Giant Panda
Toy Terrier
x
x^a